

RV32I five-stage pipeline

IF, ID, EX, MEM, WB: instruction flow, stalls, traps, and FPGA evidence

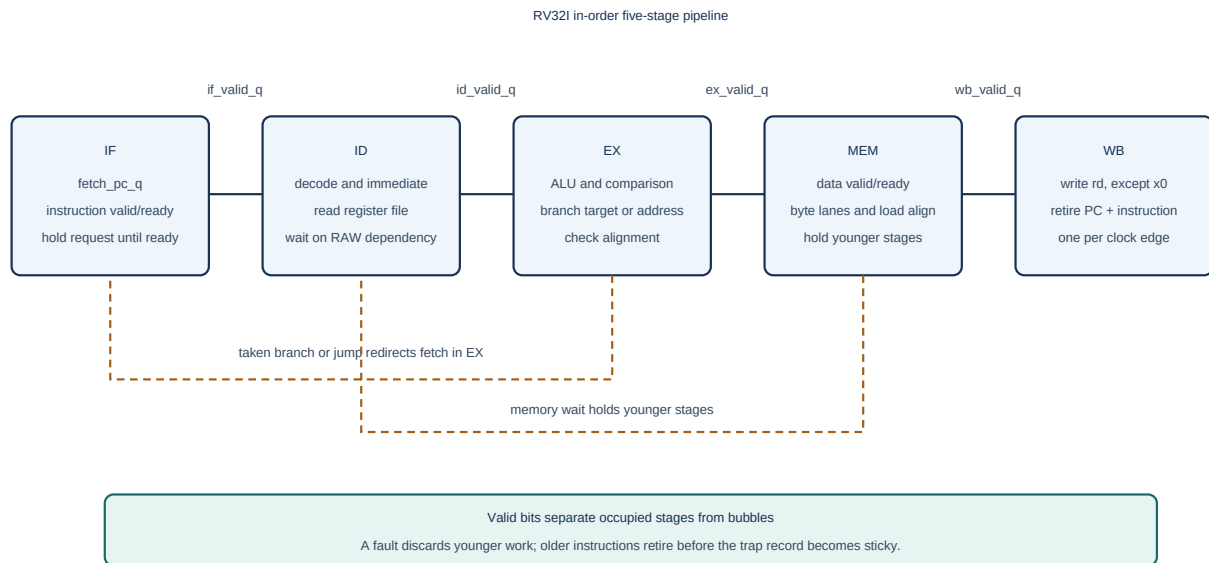


Figure 1. The valid bits mark occupied stage registers; adjacent instructions may overlap.

Design scope

The core implements unprivileged RV32I with separate ready/valid instruction and data ports. It has 32 integer registers, little-endian byte lanes, alignment checks, and a sticky external trap record. It has no forwarding, caches, privileged CSRs, interrupts, MMU, compressed instructions, or multiply/divide extension.

Pipeline registers hold an instruction's PC, data, and control decisions. A valid bit of zero marks a bubble: the nearby data bits are ignored. Unlike the earlier multicycle design, several different instructions can occupy the pipeline at once.

What each stage does

Stage	Work	Result
IF	Hold imem_valid_o and fetch_pc_q until imem_ready_i.	Instruction and PC enter IF/ID.
ID	Decode, form the immediate, and read source registers.	Operands and controls enter ID/EX when dependencies clear.
EX	Run ALU and branch comparison; form data address.	Result or memory request enters EX/MEM.
MEM	Hold a load/store request until dmem_ready_i; align load data.	Result enters MEM/WB.
WB	Write rd when legal, then pulse retirement.	Architectural register state changes.

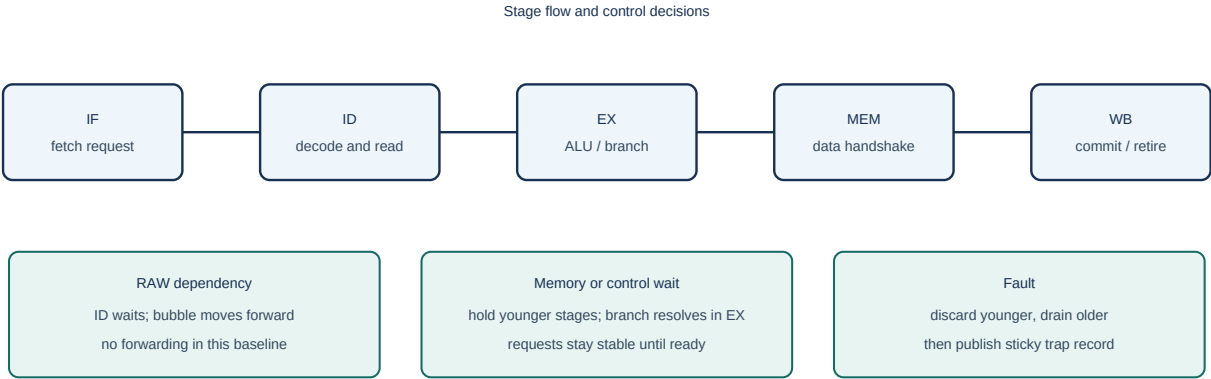


Figure 2. Pipeline flow, interlocks, and fault handling. This is not a state machine.

Reading the SystemVerilog

The suffix `_q` means a clocked value. `always_comb` describes logic that responds within a cycle. `always_ff` updates registers on a clock edge. A nonblocking assignment (`<=`) reads the old values for that edge, so neighboring instructions can advance together.

Hazards and control flow

There is no bypass network. If an instruction in ID reads a register that an older instruction will write, ID waits. A bubble moves forward until the older instruction writes in WB. For example, `ADDI x1,x0,7` followed by `ADD x2,x1,x1` waits for x1; the ADD then reads 7 and produces 14.

A waiting data request holds EX/MEM and younger stages. WB can still retire an older instruction once. Fetch pauses for an unresolved branch or jump, a memory operation, FENCE, or a fault. A branch or jump resolves in EX and supplies the next fetch PC. This simple policy limits throughput below one instruction per cycle.

Instruction and data handshakes

A request is valid while its address and controls stay stable. The matching ready input completes it. The core permits one outstanding request and pauses fetch during data-memory work. No request queue, write buffer, or cache changes the ordering.

Faults and retirement

A memory-stage fault has priority over an execute-stage fault, which has priority over a fetch fault. The faulting instruction does not retire or write rd. Younger work is discarded; older work retires before `trap_valid_o` becomes sticky. `trap_cause_o`, `trap_pc_o`, and `trap_tval_o` report the fault until reset.

`debug_state_o` is a status summary: 0 means flowing, 1 interlock or serialization, 2 memory wait, and 3 trapped. It is not a pipeline stage number.

Data operations and integration

The decoder checks opcode, funct3, and funct7. The immediate generator reconstructs I, S, B, U, and J layouts. The ALU handles arithmetic, logic, shifts, and signed or unsigned comparisons. Instruction addresses must be four-byte aligned; misaligned halfword or word data accesses trap.

Access	Legal low address bits	Write strobe
SB	00, 01, 10, 11	0001, 0010, 0100, 1000
SH	00 or 10	0011 or 1100
SW	00	1111

Loads select a byte or halfword from the returned aligned word. LB and LH sign extend; LBU and LHU fill upper bits with zeros. JAL and JALR write PC + 4 when they complete successfully. JALR clears target bit 0, and the four-byte alignment check still applies.

The SoC wrapper uses two clocked ports of a shared memory array so Vivado can infer block RAM. The FPGA top targets the Arty A7-100T. The bitstream was generated, but board execution has not been observed.

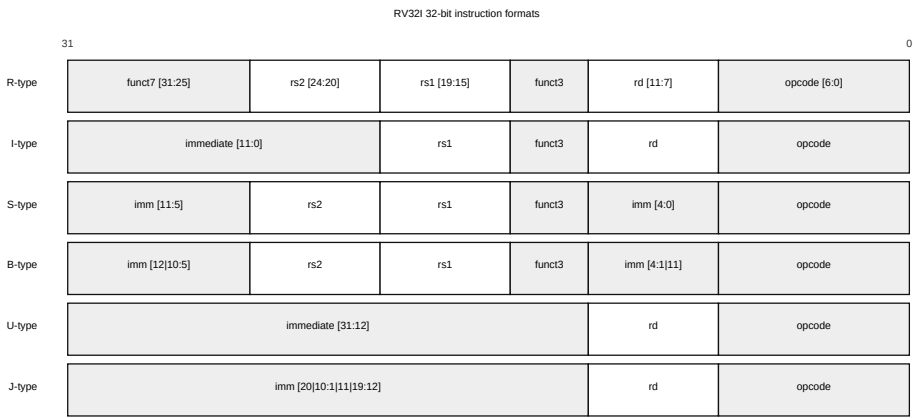


Figure 3. RV32I instruction fields. Immediate bits are reassembled by the immediate generator.

Verification and measured implementation

The cocotb environment separates a memory driver, request monitors, an independent Python ISA scoreboard, and coverage checks. This uses UVM-style roles without a SystemVerilog UVM library. Directed SystemVerilog benches check ALU operations, the instruction program, and traps.

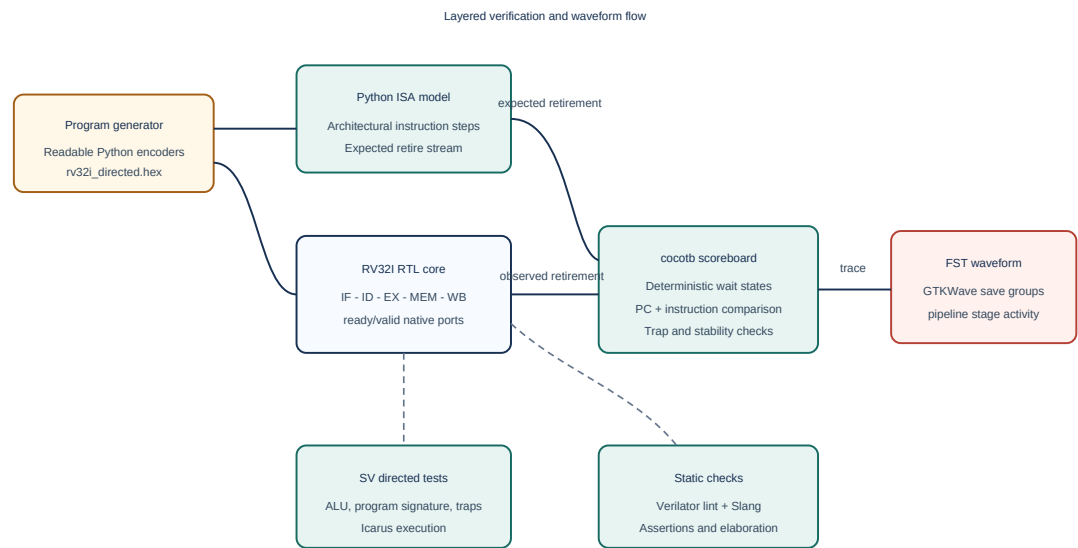


Figure 4. Checks observe the core through memory transactions and retirement outputs.

Check	Observed result
Icarus directed benches	10 ALU checks; program signature in 521 cycles; 9 traps
cocotb 2.0.1	3 tests passed, including pipeline hazards and wait states
Vivado 2023.2 XSim	ALU, directed, and trap benches passed
Arty A7-100T post-route	WNS +0.681 ns at 10 ns; 1,631 LUTs; 1,560 registers; 4 RAMB36

The implementation run had 0 DRC errors and 26 warnings. These are report results, not physical-board measurements. See docs/pipeline_vivado_report.md for warnings and build details.